

Atakan Topaloğlu

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Education

ETH Zürich | Zürich *September 2025 – Expected Graduation: June 2027*

MSc Electrical Engineering and Information Technology, GPA: 5.49/6.00

Selected Coursework: Probabilistic AI, Inference and Estimation, MPC

Koç University | Istanbul

September 2021 – June 2025

BSc in Electrical and Electronics Engineering, GPA: 3.99/4.00, Department Rank: 2/135

Selected Coursework: Digital Logic Design, C++ Programming, Signals & Systems, DSP

IEEE Signal Processing Society Scholarship - **first recipient from a university in Turkey**



Work & Research Experience

Flexion Robotics | **AI Engineering Intern**

May 2026 – Present

- Increased physics simulation **real-time factor from 0.56 to 0.90** on an RTX 5090 by eliminating unnecessary H2D syncs, reducing GIL contention, and fixing excessive collision-contact generation.
- Built an **end-to-end** sensor-agnostic LiDAR-visual-inertial **reconstruction pipeline** from raw data to sim-ready exports, **outperforming the industry baseline in mesh reconstruction** and matching 3DGS quality on small to medium sized scenes.

ETH Zurich | **Research Assistant, PRS Lab**

September 2025 – March 2026

- Redesigned point-cloud preprocessing by replacing repeated per-instance NumPy queries with a hash-based point-to-object index, reducing runtime from **8 hours to <2 minutes (~240× speedup)**.
- Diagnosed an I/O bottleneck (<1% GPU/CPU utilization), moved data augmentation offline, and added bounding-box rejection before radius-based point filtering, cutting end-to-end training from **12 days to 8 hours (~36× speedup)**.

Google | **Research Collaborator**

March 2025 – September 2025

- Developed OracleGS, achieving **state-of-the-art sparse-view Gaussian Splatting**, filtering generative priors by repurposing a regressive model as an uncertainty oracle. Open-sourced the **CUDA kernel** for depth supervision in 3DGS-MCMC.
- Led the project from proposal to publication; first author, **WACV 2026 Oral (top 3%)**.

Technical University of Munich | **Undergraduate Summer Researcher**

July 2024 – September 2024

- Repurposed a modified neural audio codec for vibrotactile data compression, achieving >30x compression ratios while preserving signal fidelity (PSNR) and real-time CPU inference.

Siemens | **Part-Time R&D Working Student**

August 2023 – July 2025

- Implemented a viewer that runs efficiently without a GPU by refactoring data layout from **AoS to SoA** to improve cache locality and enable SIMD. Implemented octrees+frustum culling to eliminate invisible geometry.
- Designed a 3D reconstruction pipeline for 3DGS, **reducing peak GPU VRAM usage by 83%** to enable processing on consumer hardware (NVIDIA Quadro P1000, 4 GB VRAM) via heavy color palette quantization and post SfM subsampling.
- Developed a wavelet-based SVM classifier to identify defects in A-Scan ultrasound data for non-destructive wind turbine testing. Secured a ~\$20k industrial kit for trial at no cost, resolving a 4-month sourcing delay.

Skills

Languages, Systems and Tools: Python, PyTorch, C++20, Docker, Nsight, Git, CMake, ONNX, SQL

Performance: CUDA, SIMD, GPU/CPU profiling

Languages (Spoken): German (Intermediate), English (TOEFL: 119/120), Turkish (Native)